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Structured assessments for imminent aggression in mental health and correctional settings: Systematic review and meta-analysis

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ABSTRACT

Background: Structured risk assessment schemes can aid violence reduction in mental health and correctional settings. However, the properties and effectiveness of schemes for predicting imminent (within 24-h) violence have not been comprehensively reviewed.
Objectives: To systematically review the properties and predictive performance of structured schemes that have been used to predict imminent aggression.
Design: Systematic review and meta-analysis.
Data sources: Multiple electronic databases were searched, relevant reference lists and journals were hand-searched.
Review methods: Thirty-two structured risk assessment schemes were identified and used in a search for evidence about the prediction of imminent violence, or their psychometric properties when used for that purpose. Study findings were synthesised narratively and, where statistics about predictive validity were available, subjected to meta-analyses.
Results: Five schemes (Brøset Violence Checklist, HCR-20: Assessing Risk for Violence version 2.0 clinical subscale, Dynamic Appraisal of Situational Aggression and its variants for youth and for women) have been subject to relevant research. Fifty papers were included in a narrative review, and 31 comprising 26 samples involving 282,238 assessments for 2723 individuals were included in meta-analyses. Only the Dynamic Appraisal of Situational Aggression and the Brøset Violence Checklist were subject to implementation research in which the tools were largely rated as having utility for, and being acceptable to, user-participants. Many studies have not established interrater reliability of the tools in the field. After removing outliers, pooled effect sizes for prediction of aggression against people and for any aggression were, respectively: Hedges' $g = 1.35[1.15–1.56]$) and $g = 1.24[0.99, 1.50]$ for the Brøset Violence Checklist and $g = 1.04$ [0.69, 1.39] and 0.88 [0.62, 1.15] for the Dynamic Appraisal of Situational Aggression. Moderator analyses revealed pooled effect sizes for both schemes were decreased by low study bias risk and increased by patient forensic vs. civil legal status.
Conclusions: Research supports the value of both the Brøset Violence Checklist and the Dynamic Appraisal of Situational Aggression. Both are generally acceptable and feasible to implement, and their predictive value is broadly equivalent producing large pooled effect sizes. There is less evidence about the Dynamic Appraisal of Situational Aggression. variants for women and youth though initial studies are promising.
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What is already known about the topic?

- Brief risk assessments for identification of violence in the near future (e.g., 24-h) in mental health settings have been developed.
- Individual studies suggest that existing tools have good predictive validity and a mixed range of other psychometric properties.

What this paper adds

- The current study identified five structured tools to assist with risk assessment for imminent violence in the coming 24-h period.
- Where calculable, pooled effect sizes across studies suggested that the best performing tool was the Brøset Violence Checklist; however, based on confidence intervals, they were comparable to those for the HCR-20: Assessing Risk for Violence clinical subscale and the Dynamic Appraisal for Situational Aggression.

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- There is evidence that those studies judged to be of better quality, those conducted in civil hospital settings, and those with higher proportions of male participants, are associated with smaller effect sizes.

1. Introduction

Violence and aggression is common in inpatient mental health and correctional facilities (Beck, 2015; Cornaggia et al., 2011). Structured, individual risk assessment is often a requirement of service providers (O'Shea et al., 2014) and research has concentrated on development of tools to predict aggression over the longer term in residential placements and following discharge or release into the community. There are two main approaches to guided violence risk assessment. First, actuarial instruments involve assignation of numerical values to a standardised schedule of known risk factors for an outcome of interest, summation of those values and comparison against norms for a comparable sample with the outcome of interest in order to make a probabilistic prediction of risk. Second, structured professional judgement schemes involve consideration of a standardised schedule of known risk factors, consideration of those risk factors in combination with clinical judgement, and knowledge of the individual. The overall level of assessed risk is usually summarised as a rating of low, medium, or high together with contextual information about the nature, severity, and likely circumstances of the outcome of interest.

The most frequently used tool globally (Singh et al., 2014), the HCR-20: Assessing Risk for Violence currently in its third version (Douglas et al., 2013) but also widely implemented in its second version (Webster et al., 1997), is a structured judgement tool that is typically used to assess risk for the coming 6- to 12-month period. Other commonly used tools include the Psychopathy Checklist Revised (Hare, 2003), an actuarial diagnostic tool for psychopathy rather than a violence risk assessment schedule, and the Violence Risk Appraisal Guide (Quinsey et al., 2006), an actuarial instrument. The latter two instruments have been used to study recidivism over periods of years (e.g., Tengstrom, 2001). The Short Term Assessment of Risk and Treatability (Webster et al., 2009), also well-used, is a structured professional judgement schedule for assessment of seven adverse outcomes including violence in the short-term (3-months). However, in correctional and mental health inpatient settings, nurses may wish to predict and take action to prevent patient violence over much shorter timespans.

1.1. Assessing risk of imminent aggression

Conceptually, risk assessment for imminent aggression has multiple potential advantages. Traditional structured judgement and actuarial tools tend to comprise historical and dynamic variables that are known, empirical risk factors for violence. However, prediction of imminent violence largely requires consideration of dynamic over historical variables since the latter are static in nature (e.g., previous violence), synonymous with hospital admission or correctional disposal, and thus of reduced potency for distinguishing patients at elevated current risk (Palmstierna and Olsson, 2007). Further, the high incidence (Cornaggia et al., 2011; Iozzino et al., 2015) and outcome severity (Tonso et al., 2016; van Leuwen and Harte, 2015) of aggression and violence speak to the need for further progress. A meta-analysis of 71 studies in mental health settings (Papadopoulos et al., 2012) reported that patient behavioural cues were immediate antecedents to 38% of violent incidents suggesting considerable scope to increase recognition of the signs of, and subsequently improve interventions to avert, imminent aggression.

1.2. Definition of imminent risk

Despite the demonstrable need, few tools have been developed which aim to aid risk assessment for imminent violence. There is no definitive period which is accepted as 'imminent'; Monahan (1981) used the term to refer to violence likely to occur in the 3-days following assessment, while other definitions include 'day-to-day risk' (Daffern, 2007), 'the next seven days' (Werner et al., 1990), 'the next few days or a week or so' (Tardiff, 2002), and 'the next two weeks' (Starzomski and Wilson, 2015). The most widely used tools to aid assessment of imminent violence, the Dynamic Appraisal of Situational Aggression (Ogloff and Daffern, 2006a) and the Brøset Violence Checklist (Almvik and Woods, 1998, 1999), are intended for rating on each nursing shift (i.e., three times a day) suggesting an intended prediction period of about 8-h. However, with rare exceptions (Almvik et al., 2007; Dumais et al., 2012), empirical studies have reported their predictive validity for the 24-h period following assessment. Accordingly, in the current paper we adopt a definition of 'imminent' as 'the next 24-h period'.

1.3. Specialist vs. non-specialist imminent violent risk assessment schemes

Adopting a definition of 'imminent risk' of 24-h highlights a conceptual difference between tools that are designed to be used for imminent violence risk prediction and those which are not but have been used anyway. Any scheme may, in principle, be used to predict imminent violence including those which are ostensibly designed for longer-term prognostication, and those which are primarily screening tools. The latter may be prioritised for completion early in an anticipated short-stay hospital admission in order distinguish which patients are at high risk over the course of that admission (e.g., Yalcin and Bilgin, 2019), or to prioritise patients for full violence risk assessment (McNiel and Binder 1994, McNiel et al., 2003; Rotter and Rosenfeld, 2018). In the current review we take an inclusive view of assessment for imminent violence in that eligibility for inclusion is based on the testing of a tool in the context of imminent (24-h) violence and not on the tools' ostensible purpose.

1.4. Contribution of the current study

Several studies have described the development, validation and predictive validity of structured tools for use in the assessment of risk for imminent aggression in forensic and civil mental health settings (Abderhalden et al., 2006; Almvik et al., 2000; Chagigiorgis et al., 2013; Daffern and Howells, 2007; Kigerl and Hamilton, 2016; Ogloff and Daffern, 2006a). However, this body of research has been subject to limited synthesis. One previous meta-analysis of imminent and non-imminent tools (Ramesh et al., 2018) only included studies conducted in forensic settings despite the use of these tools across residential mental health services. Further, the psychometric properties of the tools have been subject to limited systematic appraisal in relation to imminent violence. The aim of the present review was to identify studies which have investigated structured tools to assist in risk assessment for imminent violence (Intervention) in civil and forensic mental health, and correctional populations (Population); to evaluate the psychometric properties of the tools used including their predictive validity (Outcome). Where relevant to imminent risk, we have conducted a narrative review of results relating to user evaluations, feasibility studies and psychometric properties of the tools used, and meta-analyses of effect sizes from predictive studies to determine their performance accuracy.

2. Method

2.1. Review protocol

The study is known as the Risk Assessment Tools for Imminent Violence Evaluation Study (RATIVE Study). The protocol was not registered in advance of the study. The review comprises (i) a narrative synthesis related to the acceptability or feasibility of any tool identified as having been used for the prediction of imminent violence risk in a health or correctional setting, or any other empirical study related to its psychometric properties or its use as an intervention in the context of imminent aggression; and (ii) meta-analyses of studies of predictive efficacy where any of the tools of interest were used on multiple occasions for comparable outcomes. This report of the review followed the Preferred Reporting Items for Systematic Reviews and Meta-analyses statement (Moher et al., 2009).

2.2. Search strategy

The search was conducted in multiple computerized databases covering published, peer-reviewed literature and 'grey' literature including doctoral theses and dissertations (PsycINFO inception to present, Scopus [including Embase 1966 to present]; ProQuest [including British Nursing Database 1994 to present]; ProQuest Central [including ProQuest Dissertations & Theses Global, Social Science Database 1970 to present; Criminal Justice Database 1981 to present]; MEDLINE [inception to present]; CINAHL [1937 to present]; Academic Search Complete [1887 to present]; OPEN Dissertations [Early 20th Century to present]); Google Scholar [terms 'violence risk assessment imminent' first 100 pages]) were searched in August 2019 for articles published prior to July 2019. Additional hand-searching of references lists of studies identified through the electronic search strategy and the table of contents of relevant journals was conducted.

2.2.1. Search terms

A list of potentially relevant risk assessment tools was generated based on the authors' knowledge of the field and on examination of relevant systematic reviews or surveys of the use of violence risk assessment instrumentation (Hurducas et al., 2014; Ramesh et al., 2018; Singh et al., 2014). This was supplemented, in a second iteration of the main search, by additional tools identified in the first search iteration. Thirty-two instruments, including multiple versions, were each included (see example search Table 1) as a search-term using its acronym and full name together with the family name of the first author (see Table 1). Results were combined with terms related to population and intervention domains. All searches of electronic databases were conducted using the "anywhere in the text" option where it was available. Titles and abstracts of search results were examined to determine their suitability for inclusion. In cases of doubt, the full text was acquired (including contact with the corresponding author). All assessment tools used to predict imminent aggression were noted and subject to further searches (tool name and author but without further limiters).

2.2.2. Inclusion and exclusion criteria

In order to be included in the narrative review a paper must have described an empirical account of user evaluations, feasibility studies, psychometric properties, or predictive validity of a structured risk assessment tool for imminent (24-h) violence. To reiterate, studies must have been relevant to imminent violence risk assessment; thus, for example, studies of the HCR-20: Assessing Risk for Violence tool would only be included where the results were explicitly about its use as an imminent violence risk assessment tool. For inclusion in the meta-analyses the paper must have described an empirical account of a study of predictive validity of any of the identified assessment tools for any violent outcome occurring within 24-h of assessment in any population. Further, an area under the curve (AUC) value, or sufficient statistical information to calculate such a value, was required. Exclusion criteria for the narrative review were non-primary research (i.e., reviews), and non-English language papers. Exclusion criteria for meta-analyses were: papers where the assessment instruments used were scored substantially differently from the authors' recommendation including retrospective assessment (Loi and Marlowe, 2017), or by machine algorithm (Hotzy et al., 2018; Van Le et al., 2018); and studies or portions of studies where prediction took place for a period of >24-h following risk assessment (e.g., Meehan et al., 2017; Vaaler et al., 2011). Where multiple articles contained overlapping samples, only the study with the largest sample size was retained to avoid including the same participants twice; however, if different tools or outcomes were examined, both were retained.

2.2.3. Tools identified as relevant in the search

Studies involving one or more of five instruments were identified from the search as having been subject to imminent violence-related research. The five instruments comprised a total of 21 unique items of which 14 were included in one tool only. Of the remainder, three items were shared by five tools, two by four, and two by three.

Brøset violence checklist (Almvik and Woods, 1999): Based on an empirical investigation of common antecedents to inpatient aggression against others (Linaker and Busch-Iversen, 1995), the Brøset Violence Checklist comprises six items: confusion, irritability, boisterousness, physical threats, verbal threats, and attacking objects. Each is rated as present (score 1) or absent (score 0) since the most recent assessment. If the item always applies it is only rated present when there is a qualitative change in its nature or intensity. The tool should be completed on a per shift basis (night/ day/ evening). The ratings are summed and used actuarially: a total score of 0 indicates that the risk of violence is small; 1 or 2 indicates that risk is moderate and requires preventative measures; and a score of 3+ indicates high risk, a need for preventative measures, and scenario-planning. Suggested interventions are verbal de-escalation, diversional activity, stimulation, sensory modulation, medication, continuous supervision, seclusion, and restraint.

HCR-20: assessing risk for violence version 2.0 clinical subscale (Webster et al., 1997): The 20-item HCR-20: Assessing Risk for Violence tool is recommended for assessment for violence in the coming 6–12 months both in institutional and community settings. The five clinical subscale items are lack of insight, negative attitudes, active symptoms of major mental illness, impulsivity, and non-responsivity to treatment. Each is rated on a 3-point scale as not present, present to some extent, or definitely present. In some studies of imminent aggression, for the stated purpose of ensuring consistency with the dichotomously rated Dynamic Appraisal of Situational Aggression, the 3-point scoring system has been amended to a Yes/No scale for rating (Daffern and Howells, 2007; Daffern et al., 2009) or has been collapsed to a dichotomous rating (definitely present or present to some extent vs. not present) prior to analysis (Chu et al., 2013a, 2013b).

Dynamic appraisal of situational aggression (Ogloff and Daffern, 2006): The Dynamic Appraisal of Situational Aggression comprises two Brøset Violence Checklist items (irritability and verbal threats), two HCR-20: Assessing Risk for Violence clinical subscale items (impulsivity and negative attitudes), and three of the developers' own items (sensitive to perceived provocation, easily angered when requests denied, and unwillingness to follow directions). Each of the seven items is rated for its

Table 1
Example literature search (Database: Scopus [Includes Embase]).

#	Search term	*N* Results
1	((bvc OR brøset AND violence AND checklist) AND (woods OR almvik)) OR ((elm-b OR (east AND London AND modified AND broset) AND Loi))	
2	((covr OR classification AND of AND violence AND risk) AND (monahan))	
3	((dasa OR dynamic AND appraisal AND of AND situational AND aggression) AND (ogloff OR daffern))	
4	((dasa wv OR dynamic AND appraisal AND of AND situational AND aggression AND women) AND (ogloff OR daffern))	
5	((dasa yv OR dynamic AND appraisal AND of AND situational AND aggression AND youth AND version) AND (ogloff OR daffern))	
6	((brc OR brockville AND risk AND checklist) AND (chagigiorgis))	
7	((fotres OR forensisch AND operationalisiertes AND therapie- AND und AND risiko-evaluations-system) AND (urbaniok))	
8	fordham AND risk AND screening AND tool) AND (rosenfeld))	
9	((frst) OR (ALL ((harm AND fv OR hamilton AND anatomy AND of AND risk AND management AND forensic AND version) AND (cook))	
10	((hcr-20) OR (historical AND clinical AND risk AND management-20) AND (webster)))	
11	((hcr-20 AND version 3 OR historical AND clinical AND risk AND management-20 AND version 3) AND (douglas))	
12	((hcr-20 AND (female AND additional AND manual) AND (de vogel))	
13	((hkt-30 OR historische, AND klinische, AND toekomstige-30) AND (werkgroep AND pilotstudy AND risicotaxatie))	
14	((iorns OR inventory AND of AND offender AND risk, AND needs AND strengths) AND (miller))	
15	((irrs OR imminent AND risk AND rating AND scale) AND (starzomski OR wilson))	
16	((lsi-r OR level AND of AND service AND inventory-revised) AND (andrews OR bonta))	
17	((pcl OR psychopathy AND checklist) AND (hare)))	
18	((pcl-sv OR psychopathy AND checklist-screening AND version) AND (hare)))	
19	((pcl-r OR psychopathy AND checklist-revised) AND (hare))	
20	((saprof OR structured AND assessment AND of AND protective AND factors) AND (de AND vogel))	
21	((sara OR spousal AND assault AND risk AND assessment AND guide) AND (kropp))	
22	((savry OR structured AND assessment AND of AND violence AND risk AND in AND youth) AND (borum))	
23	((sorag OR sex AND offender AND risk AND appraisal AND guide) AND (quinsey)))	
24	((start OR short-term AND assessment AND of AND risk AND treatability) AND (webster))	
25	((static-99 OR static-99r OR static-2002 OR static-2002r) AND (hanson OR thornton))	
26	((svr-20 OR sexual AND violence AND risk-20) AND (boer))	
27	((v-risk-10 OR violence AND risk AND screening-10) AND (hartvig))	
28	((vrag OR violence AND risk AND appraisal AND guide) AND (quinsey))	
29	((vrs OR violence AND risk AND scale) AND (wong OR gordon))	
30	((vsc OR vsc-r OR (violence AND screening AND checklist) AND (binder OR mcniel))	
31	((violence AND screening AND checklist and revised) AND (mcniel OR binder))	
32	1 OR 2 OR 3 OR 4 OR 5 OR 6 OR 7 OR 8OR 9 OR 10 OR 11 OR 12 OR 13 OR 14 OR 15 OR 16 OR 17 OR 18 OR 19 OR 20 OR 21 OR 22 OR 23 OR 24 OR 25 OR 26 OR 27 OR 28 OR 29 OR 30 OR 31	26,305
33	((aggress* OR violen* OR assault OR battery OR attack OR assail OR threat* OR hostil* OR abuse))	351 863
34	((inpatient OR hospital OR prison OR gaol OR gaol OR correctional* OR penitentiary OR maximum AND secur* OR medium AND secur* OR high AND secur* OR low AND secur* OR (ward AND acute)))	239,549
35	((prediction OR predictive OR predictive AND validity OR recidivism OR reoffend*))	355 427
36	((imminent OR short AND term OR 24 h OR 24-h OR one AND day OR immediate OR impending))	315,992
37	33 OR 34 OR 35 OR 36	4 351 863
38	32 AND 37	395

presence or absence in the preceding period. The tool's manual (Ogloff and Daffern, 2006a) suggests scores of 0–1 represent low risk requiring no remedial action; scores of 1–2 represent moderate risk requiring consideration of preventative action, closer monitoring, and communication of increased risk to the clinical team; and scores of 3+ require remedial action and a risk management plan. The tool was developed, the authors state, because the Brøset Violence Checklist required more clinically relevant items that reflect treatment-amenable, modifiable risk factors (Ogloff and Daffern, 2006b).

Dynamic appraisal of situational aggression: youth version (Daffern et al., 2008): This version of the Dynamic Appraisal of Situational Aggression comprises 11 items, all seven from the original tool plus anxious or fearful, low empathy/remorse, significant peer rejection, and outside stressors and all scored similarly to the parent tool.

Dynamic appraisal of situational aggression: women's version (Riordan et al., 2019): This version of the Dynamic Appraisal of Situational Aggression comprises 10 items, seven from the original tool plus three new items: two female-specific clinical items from the HCR-20: Assessing Risk for Violence Female Additional Manual (de Vogel et al., 2012, 2014): covert/manipulative behaviour and low self-esteem; and a binary social climate item.

Tools with no associated imminent risk studies: Studies involving nine tools were not included because, while those tools' stated aim was to aid assessment of prediction for imminent or short-term violence, the period examined in studies was longer than 24-h. They were the Brockville Risk Checklist (Chagigiorgis et al., 2013); the Hamilton Anatomy of Risk Management–Forensic Version (Cook et al., 2018); the Violence Risk screening 10 (Hartvig et al., 2011); the Fordham Risk Screening Tool (Rosenfeld et al., 2017); the Imminent Risk Rating Scale (Starzomski and Wilson, 2015); the Short Term Assessment of Risk and Treatability (Webster et al., 2009); the Violence Screening Checklist and its later revised version (McNiel and Binder 1994, McNiel et al., 2003); the HCR-20: Assessing Risk for Violence Version 3.0 (Douglas et al., 2013) including its summary risk rating for imminent violence. The East London Modified-Broset (Loi and Marlowe, 2017), an adapted Brøset Violence Checklist with two additional items and definitional clarifications of the remaining six, aimed to aid decision-making for seclusion and was excluded. Studies using the remaining 18 tools, none designed to be used for imminent violence risk assessment, did not include any examination of imminent violence-related properties.

2.3. Data extraction

Study details were extracted (see Table 2) including relevant results about (a) user evaluations of feasibility and utility; (b) face, content, or convergent and concurrent validity; (c) interrater reliability; (d) predictive validity (including differential predictive validity); and (e) incremental validity. Statistics describing or facil-

Table 2
Characteristics of studies included in the meta-analyses.

Study	Country	Setting	Sample size (% PP†)	*N* Assess-ments (% PP†)	Design	Mean age (Years)	Gender	Diagnosis	Risk Tool(s) and Outcome	Follow up	QUADAS-2 Bias risk	Rated by
Ogloff & Daffern (2006)	Australia	Forensic. Secure hospital	100 (50.0)	10,043 (1.11)	Prospective	33.0	Mixed (78% M)	Psychosis (77%); mood disorders (11%)	DASA C BVC PP	24-h	At risk Domain 2	Nurses
Daffern and Howells (2007) / Daffern et al. (2009)	UK	Forensic. High secure PD hospital	44 (45.5 Any)	1223 (6.95 Any)	Prospective	34.4	M	Antisocial PD (72.7%); Borderline PD (40.9%); Narcissistic PD (15.9%)	DASA C SS, SH	24-h	At risk Domain 2	Nurses
Barry Walsh et al. (2009)	New Zealand	Mixed. Locked, low-secure	58 (–)	10,013 (0.88)	Prospective	36.1	Mixed (96.6% M)	Schizophrenia (46.6%); Bipolar (17.2%); intellectual or developmental disability (36.2%)	DASA PO, VA PP, AA	24-h	At risk Domain 2	Nurses
Vojt et al. (2010; 2013)	UK	Forensic. High secure hospital	26 (30.8)	3640* (2.5)	Pros-pective	36.0	M	Psychosis (77%)	DASA VA PP AA	24-h	At risk Domain 2, 3	Nurses
Dumais et al. (2012)	Canada	Civil. PICU	77 (10.4)	3798 (1.10)	Prospective	36.4	Mixed (61.0% M)	Schizophrenia or schizoaffective disorder (48.7%); bipolar (34.2%); depression (6.6%)	DASA PO, PP	8-h/ 24-h	At risk Domains 2, 3	Nurses
Chu et al. (2012)	Singapore	Correct-ional/Foren-sic Youth institutions	49 (34.7 Any)	2008 (1.29)	Prospective	15.4	Mixed (59.2% M)	No data	DASA VA, PP AA	24-h/ 48-h	Low Risk	Staff
Chu et al. (2013a, 2013b)	Australia	Forensic. Secure hospital	70 (22.9)	3449 (1.22)	Prospective	34.3	Mixed (78.6% M)	Multiple comorbidities (40%); psychosis (80.0%); substance use (74.3%); PD (20.0%); affective(11.4%)	DASA C BVC VA, PP, AA	24-h	At risk Domain 2	Nurses
Griffith et al. (2013)	Australia	Civil. Acute/ secure MH hospital	– (–)	1230 (0.22 Any)	Prospective	–	–	No data	DASA PP	24-h	Low Risk	Nurses
Chan & Chow (2014)	China	Forensic. Max. secure psychiatric unit	530 (17.7 PP PO)	7359 (4.93 PP PO)	Prospective	39.2	Mixed (76.4% M)	Schizophrenia and related (30.0%); substance misuse (22.4%)	DASA, BVC PP	24-h	Low Risk	Staff
Kasinathan et al. (2015)	Australia	Forensic. Adolescent high secure unit	35 (–)	4440 (1.37)	Prospective	–	Mixed (97.1% M)	Schizophrenia (83.0%); bipolar (11.0%); conduct disorder (89.0%)	DASA DASA:YV VA, PP, PO, AA	24-h	At risk Domain 2	Nurses
Maguire et al. (2017; 2018)	Australia	Forensic. Secure hospital	60 ‡ (92.0 Any)	3600 (1.28 PP)	Retrospective file review	–	Mixed (50.0% M)	Schizophrenia (70%); other schizophrenia spectrum (21.7%); major depression (3.3%); PD (3.3%); bipolar (1.7%)	DASA PP	24-h	At risk Domain 2	Nurses
Lantta et al. (2016)	Finland	3 civil inpatient mental health units	12 (8.3)	178 (1.69)	Prospective	–	–	Schizophrenic disorder (42.0 – 64.0% across sites)	DASA VA, PP	24-h	At risk Domain 1, 2	Nurses
Nqwaku et al. (2018)	UK	Forensic High secure mental health unit	43 (–)	3564 (0.42 PP)	Prospective	–	M	Primary diagnosis i) mental illness; ii) PD; iii) mental illness and PD	DASA AA	24-h	At risk Domain 2	Nurses
Riordan et al. (2019)	Australia	Forensic High Security mental health hospital	13 (–)	13,442 (7.7)	Prospective	39	F	Psychosis (92%); PD (77%); Intellectual disability (15%)	DASA, DASA:WV VA, PO, PP, SH, AA	24-h	At risk Domain 2	Nurses

(*Continued on next page*)

Table 2 (*Continued*).

Study	Country	Setting	Sample size (% PP†)	*N* Assess-ments (% PP†)	Design	Mean age (Years)	Gender	Diagnosis	Risk Tool(s) and Outcome	Follow up	QUADAS-2 Bias risk	Rated by
Dutch and Patil (2018)	USA	8-bed locked acute in-patient psychiatric unit	83 (–)	465 (4.3)	Prospective	Mdn 14	Mixed (33.3% M)	No data	DASA, DASA:YV VA, PO, PP, AA	24-h	Low Risk	Nurses
Abderhalden et al. (2004; 2006) Rechen-macher et al. (2014)	Switzerland	Civil. Six acute admission wards	219 (4.6)	1203 (1.16)	Prospective	39.8	Mixed (60.3% M)	Schizophrenia (34.9%); Alcohol/ Substance use disorder (23.9%); affective (14.8%)	BVC PP	Next shift	At risk Domain 2	Nurses
		Civil. Two wards from 2 different hospitals	300 (9.0)	2084 (1.78)	Prospective	36.4	Mixed (61.0% M)	Schizophrenia (40.1%); Substance use disorder (22.8%); affective (16.5%)	BVC PP[c]	Next shift	Low Risk	Nurses
Almvik et al. (2000) Almvik & Woods (1998; 1999)	Norway	Civil. Acute inpatient wards	109 (11.0)	901 (3.66)	Prospective	Mdn 30 - 40	Mixed (47.7% M)	No data	BVC PP	24-h	At risk Domain 2	Nurses
Almvik et al. (2007)	Norway	Civil psycho geriatric / dementia wards	82 (–)	8896 (2.38 Any)	Prospective	82.0	Mixed (43.9% M)	Dementia (78.1%); depression (12.2%); bipolar (3.6%); psychosis (6.1%)	BVC PP	8-h/ 24-h	At risk Domain 2	Nurses
Hvidhjelm et al. (2014)	Denmark	6 low/med-ium/high secure for-ensic wards	156 ‡ (48.7 PP/PO)	139,579 (0.3 PP/PO)	Prospective	38.4	Mixed (96% M)	Schizophrenia (83.3%); affective (3.8%)	BVC PP	24-h	At risk Domain 2	Staff
Woods et al. (2015)	Canada	Forensic. Secure unit	46 (–)	2492 (2.05 Any)	Prospective	35.9	Mixed (76.1% M)	Psychosis (58.7%); substance misuse (91.3%); affective (13.0%); dementia (4.3%)	BVC PP	Same shift	At risk Domain 2	Nurses
Yao et al. (2014)	China	Civil. Two acute psychiatric wards	281 ‡ (19.6 Any)	3707 ‡ (1.35 Any)	Prospective	34.5	Mixed (43.1% M)	Schizophrenia spectrum (70.0%); affective (23.0%)	BVC PP	Interval be-tween daily ratings	Low Risk	Nurses
Marques et al. (2015)	Portugal	Civil. Four short stay psychiatric units	64 (†20.3 Any)	64 (21.88 PP PO)	Prospective	44.0	Mixed (43.7% M)	Schizophrenia spectrum 43.8%	BVC AA	24-h	At risk Domain 2	Nurses
Moursel et al. (2018)	Turkey	Adult psychiatric clinic of a university hospital	126 (†19.8 Any)	504 (–)	Prospective	–	Mixed (50..0% M)	Depression 36.5%; Bipolar 30.9%; Schizophrenia spectrum 21.4%; PD 4.8%	BVC AA	24-h	Low Risk	Primary nurse/ care-giver
Lansgrud et al. (2019)	Norway	Civil. PICU	40 (†15.0)Any)	443 († Any 3.2)	Prospective	41.7	Mixed (58.0% M)	Schizophrenia spectrum 42.0%, Mood disorders 26.0%; Substance 18.0%	BVC AA	24-h	At Risk Domain 2	Nurses

Abbreviations: BVC Brøset Violence Checklist; DASA Dynamic Appraisal of Situational Aggression (YV Youth Version; WV Women's Version); C HCR-20: Assessing Risk for Violence clinical subscale; PP Physical Aggression Against People; PO Physical Aggression Against Objects; Any – Any Aggression; VA Violent/Aggressive; PD Personality Disorder; M Male; QUADAS-2 (Quality Assessment of Diagnostic Accuracy Scale version 2, Whiting et al., 2011). Domain 1 = Potential risk of bias due to patient selection; Domain 2 = Potential risk of bias from conduct of index test [in all cases here from lack of detail about inter rater reliability]; Domain 3 = Potential risk of bias due to concerns about the reference standard [i.e. violence outcomes measurement]; Domain 4 Potential risk of bias due to flow and timing of study [e.g., dropouts, intervals in data collection]. †PP unless otherwise stated.

itating calculation of predictive accuracy (Area Under the Curve)) were extracted. Further, because the goal of risk assessment is to prevent and reduce aggression and violence, we also extracted results where studies examined the ability of any included tool to (g) detect reliable change over time or (h) establish the effectiveness of tool use as an intervention.

2.4. Risk of bias

The quality of studies included in the meta-analyses was assessed independently by two authors (A1 and A3) using the Quality Assessment tool for Diagnostic Accuracy Studies version 2 (Whiting et al., 2011). Rating occurs across four domains (1. patient selection; 2. conduct and interpretation of the index test; 3. conduct and interpretation of the reference standard; and 4. study flow and timing) for risk of bias (high/low/unclear) and concerns regarding applicability (high/low/unclear). The overall risk of bias was coded based on the judgement of all domains as 'low' (overall low risk of bias) or as 'at risk' (based on rating of risk or unclear risk on one or more domains). This produced a Kappa of 0.90, and the single discrepancy was resolved through discussion. All eligible studies were included in the meta-analyses irrespective of bias.

2.5. Data synthesis

2.5.1. Narrative review

A narrative synthesis was conducted guided by recommendations made by the UK Economic and Social Research Council (Popay et al., 2006). Almost all data were quantitative in nature, though only outcomes related to predictive efficacy were suitable for meta-analyses. Study findings were extracted and grouped into clusters representing categories related to characteristics of good psychometric tools. Information was tabulated and descriptive statistics (frequencies, means) were calculated about study and participant characteristics. Results were examined using vote counting techniques and elements of thematic analysis. Initial and subsequent drafts of the narrative synthesis were circulated among the study team; as a result, suggestions for amendments to clusters and interpretations of results were incorporated iteratively. AUC*s* representing predictive efficacy, each comprising a score representing a predictor variable (risk scale used) and an outcome (e.g., aggression type), were organised into clusters in terms of the tool used and then outcome type and, where relevant, target (e.g., physical aggression against people, physical aggression against objects). Because some studies examined the predictive efficacy of multiple tools for the same outcomes in a single sample we clustered data by tool in order to avoid pooling data from the same sample. As a result, we did not calculate a pooled effect size for all tools for a specific outcome. Where practicable, more specific categories of aggression (e.g. specified type *and* specified target) were pooled into higher order categories (e.g., specified type, *no* specified target). Thus, where outcomes were reported separately for physical aggression against patients and for physical aggression against staff, but not for physical aggression against people, the two separate outcomes effect sizes were treated as a "person-directed physical aggression" outcome and were pooled with results from studies reporting a physical assault against people outcome. The final outcome categories were: "any aggression", "physical aggression against people", "physical aggression against objects"', "verbal aggression", and "self-harm". All outcomes were defined and classified by the individual study authors; therefore, there may be some variation in the operational definitions used across studies but clustered together in our analyses.

2.5.2. Meta-analyses

AUC values were converted to Cohen's *d* values, which are a commonly used measure of effect size appropriate for random effects models (Yang et al., 2010), using an online conversion spreadsheet (www.stat-help.com). Since Cohen's *d* values can be biased and overestimate the true effect size for small sample sizes (Lakens, 2013) they were converted to more conservative Hedge's *g* values using formulas provided in the compute.es package (Del Re, 2013). Hedges' *g* values are interpreted similarly to Cohens *d*: i.e., small effect = 0.2; medium effect = 0.5; large effect = 0.8 (Cohen, 1977; Hedges and Olkin, 1985).

Meta-analyses of predictive validity studies was conducted using R (R Core Team, 2015). Hedge's *g* values, weighted by inverse variance, were pooled in a random-effects model using the rma function in the metafor package (Viechtbauer, 2010). This calculation requires entry of the standard error (SE) of study effect sizes: given that prediction was for events rather than individuals, these were calculated based on the dichotomous outcome of all events (aggressive vs. non-aggressive) in the sample. In addition, results from a fixed effects model were calculated because random effects models can be unstable when there are <30 studies per analysis (Schulze, 2007). Where both models provide similar results then greater confidence can be invested in the findings. The metaphor package calculates 95% confidence intervals and conducts homogeneity tests using the Q and I^2 statistics. The former is distributed as a chi-square with k–1 degrees of freedom (where $k = n$ independent effect sizes) and the latter describes the percentage of variation across studies that is due to heterogeneity rather than chance (Higgins and Thompson, 2002; Higgins et al., 2003). $I^2 = 100\% \times (Q - \text{d.f.})/Q$.

In order to detect outlier cases that could overly influence the study data set, we used methods suggested by Hanson and Bussiere (1998). Forest plots were visually inspected; potential outliers were defined as individual studies whose effect size 95% CIs fell outside those of the pooled effect size of all studies in that analysis. The analysis was then repeated without the potential outlier and, where the subsequent Q statistic was half or less of its original magnitude, the removed study was defined as an outlier. Changes in both fixed and random effect pooled estimates from the analyses with and without outliers were noted.

2.5.3. Moderator analyses

Where significant heterogeneity was detected in meta-analyses subsequent to removal of outliers then moderator analyses were conducted to examine the source of variance. Due to the small number of studies includable in the moderator analyses the number of variables investigated was limited to five. First, gender, coded as the proportion of the sample that was male, was included because previous research shows that risk assessment schemes including the HCR-20: Assessing Risk for Violence and the Short Term Assessment of Risk and Treatability perform more accurately in women (O'Shea and Dickens, 2015; O'Shea et al., 2013). Second, legal status (civil, forensic or mixed containment: reference category civil) was included since it is reasonable to assume that prediction will improve in the more stable, longer term environments found in forensic services. Third, overall study bias (reference category: Low) was included to investigate the effect of bias on estimated effect size. Fourth and fifth were study level variables: violence base rate calculated as the number of aggressive incidents as a proportion (%) of all opportunities for aggression (number of observations); and the number of observations per participant. This last potential moderator was included because studies of predictive validity for imminent aggression are unusual in the violence risk assessment literature for their inclusion of multiple ratings per participant.

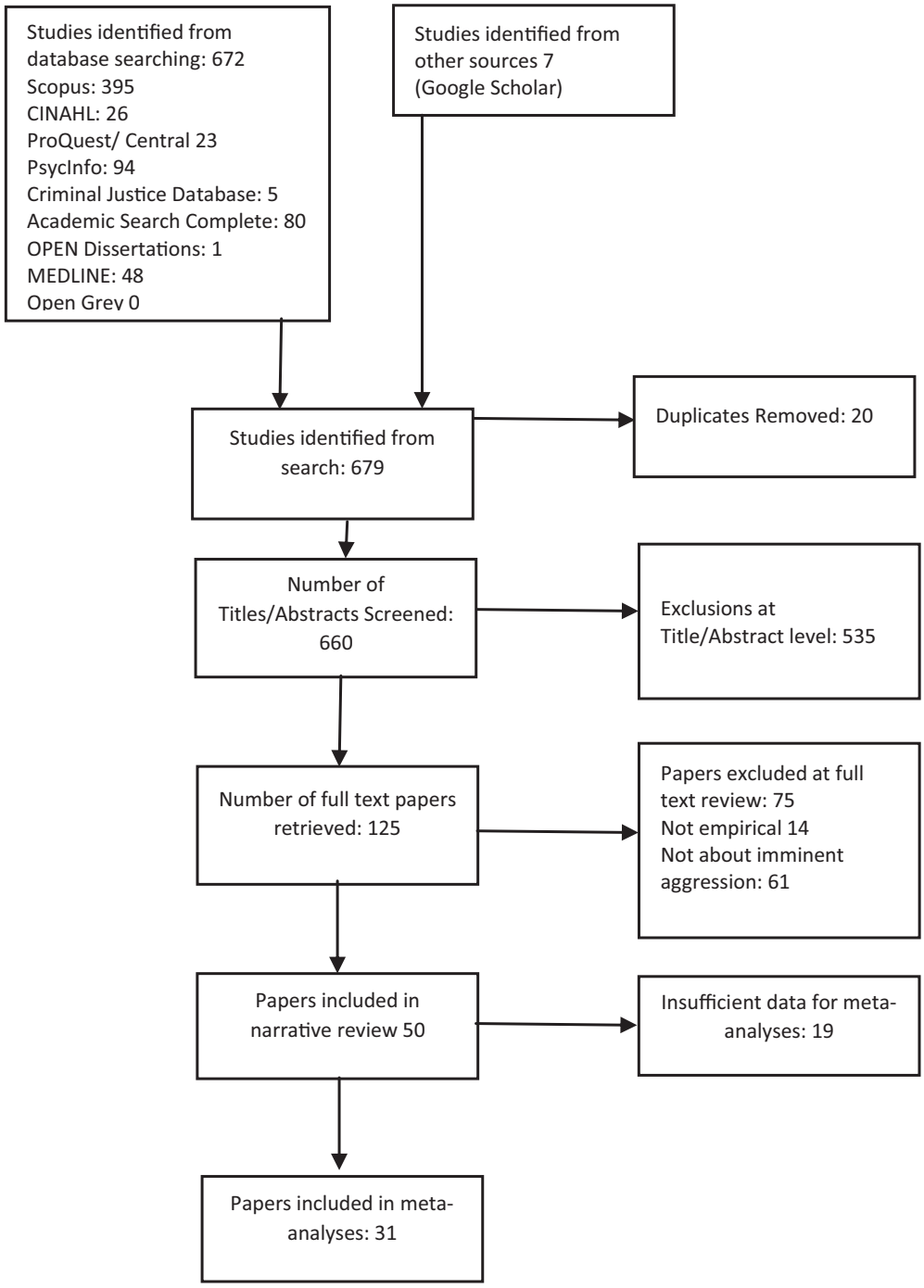

Fig. 1. Preferred reporting items for systematic reviews and meta-analyses (Moher et al., 2009) study selection flow chart.

3. Results

3.1. Study characteristics

The search strategy identified 679 records, of which 50 were included in the narrative review and 31 were eligible for the meta-analyses (see Fig. 1). These 31 papers included 25 unique samples (one sample was covered in two different papers on three occasions, and in three different papers on one occasion; a two-sample study was reported in three different publications) comprising 282,238 risk assessments for a total of 2723 individuals and 228,325 incident opportunities (i.e., because some patients were assessed with more than one tool per study there were more risk assessments than risk opportunities). Of these, Dynamic Appraisal of Situational Aggression – Outcome and Brøset Violence Checklist – Outcome pairings were examined in 14 and 13 samples respectively; the Dynamic Appraisal of Situational Aggression Youth Version and HCR-20: Assessing Risk for Violence clinical subscale were each examined in two studies, and the Dynamic Appraisal of Situational Aggression Women's Version in one. Four studies examined >1 tool and, of these, two studied three tools each. The Dynamic Appraisal of Situational Aggression sample comprised $n = 1117$ ($M = 85.9$) individuals, the Brøset Violence Checklist sample $n = 2079$ ($M = 172.8$); $n = 170$ individuals were studied using

the HCR-20: Assessing Risk for Violence clinical subscale, and 118 using the Dynamic Appraisal of Situational Aggression Youth Version; $n = 735$ individuals were included in analyses of>1 tool. Participants in the included samples were predominantly male ($M = 71.8\%$ SD=22.6); mean age was 38.3 years (SD=13.5, range 15.4–82.0). All studies involved adults except those by Chu et al. (2012) where mean age was 15.4 years; and Kasinathan et al. (2015) where age range was 15–21 years. Almvik et al. (2007) studied older age adults ($M = 82$ years). Samples were drawn from 14 nations; Australia ($k = 6$) and the UK and Norway ($k = 3$ each) were the most commonly represented. Twelve samples each came from forensic and civil settings and one from a mixed setting. Risk of bias was judged to be present in studies involving 17 of the 25 samples (68.0%); most common sources of potential bias was lack of information regarding risk measurement, particularly the lack of reporting about robust and reliable procedures for data collection by multiple assessors (see Table 2).

3.2. Narrative review

3.2.1. Feasibility and utility

Fourteen papers evaluated the implementation, feasibility and/or utility of the Dynamic Appraisal of Situational Aggression or the Brøset Violence Checklist. No study examined these properties for the other tools.

Brøset violence checklist: Reported time-to-complete a single assessment was 1 to 3 min (Abderhalden et al., 2004; Clarke et al., 2010; Rechenmacher et al., 2014). Respondents frequently reported items to be easy to understand (Abderhalden et al., 2004; Clarke et al., 2010), user-friendly (Woods et al., 2008; Yao et al., 2014), and useful in at least some contexts (Woods et al., 2015). However, respondents did not always feel that the tool improved their awareness of potentially aggressive patients (Clarke et al., 2010). One of six respondents in Clarke et al. (2010) found some terms ambiguous and one user in Woods et al. (2015) said that it largely duplicated current, routinely-collected data. In a longer-term evaluation, Clarke et al. (2010) reported that, 1-year post-implementation, 83% of respondents on the implementation unit said the tool was easy to use and 67% were in favour of continued use. At 5-year follow-up, however, an initial expansion of tool use across the entire 88-bed service had contracted to use on two units.

Dynamic appraisal of situational aggression: Alford-Jenkins' (2014) small survey of nurses ($N = 3$) conducting violence risk assessments for 7-days in a male psychiatric unit reported that respondents found the tool easy to use and wanted to continue use. Lang's (2015) survey of correctional officers using the Dynamic Appraisal of Situational Aggression in four New Zealand prisons revealed 61% found it very or moderately easy to administer and 35.5% neither easy nor hard. Most agreed it improved transfer of information and half indicated that it improved their knowledge of signs of imminent aggression. Underwood (2017) examined nurses' ($N = 20$) responses to a 5-item survey following the introduction of the tool into an emergency psychiatric setting. Most agreed that using the it had helped them identify potentially aggressive behaviour (90%); increased their awareness of violent behaviour (85%); had triggered them to make an intervention to prevent aggression (85%); had reduced use of seclusion and restraint (75%); and 70% wanted to continue using the tool. In contrast, Daffern et al. (2009) reported that nurses involved in the tool's implementation disagreed that structured outperformed unstructured clinical judgement and they did not routinely incorporate resulting information into the nursing clinical handover. Dumais et al. (2012) reported that >75% of participants agreed that the tool was relevant, useful, and was used to communicate clinical information to other professionals.

Lantta and colleagues (Lantta, 2016; Lantta et al., 2015; 2016) have described and evaluated Dynamic Appraisal of Situational Aggression implementation in a Finnish psychiatric setting. Potential barriers to implementation identified by nurses included motivation to use the tool, counter-productive outcomes if unmotivated nurses use the tool carelessly, and lack of certainty about using resulting assessment data to inform interventions. Potential facilitators included encouragement, management 'buy-in', use of checklists to improve adherence, a list of recommended interventions, and use of results routinely in nursing care summaries. Staff feedback following implementation was positive (quick to use, precise, increased interactions, suitability) and negative (preference for intuition, unsuitability, prevention and assessment hard to link, lack of training); patients also provided feedback that was positive (increased communication; interest) and negative (negative focus; concerns about objectiveness; time consuming; lack of opportunity to discuss assessments and their consequences).

3.2.2. Face and content validity

Clarke et al. (2010) found that the Brøset Violence Checklist 'somewhat reflected' patients' behaviour. No similar information has been presented for the remaining tools.

3.2.3. Reliability

Yao et al. (2014) reported intraclass correlation coefficients for inter-rater reliability of 0.41–0.76 for Brøset Violence Checklist items (Mdn=0.63) and 0.84 for total score suggesting good performance. Moursel et al. (2018) reported an interrater reliability coefficient (Cohen's $K = 0.93$) based on two raters' independent Brøset Violence Checklist ratings of 30 patients. Chan and Chow (2014) reported that rating of two practice Dynamic Appraisal of Situational Aggression assessments from videotaped hypothetical vignettes produced a Krippendorff's alpha of 0.92 suggesting excellent interrater reliability. Following Dynamic Appraisal of Situational Aggression training, participants in Chu et al., 2012 study rated two clinical vignettes each and achieved an intraclass correlation coefficient of 0.91. Dutch and Patil (2018) reported a Kappa of 0.79 (95% CI 0.70, 0.89) for staff pairs' ratings of 47 Dynamic Appraisal of Situational Aggression Youth Version assessments from 24 patients.

3.2.4. Convergent and concurrent validity

Dynamic Appraisal of Situational Aggression and Brøset Violence Checklist total scores were highly correlated in studies by Chan and Chow (2014) (Spearman's r_s=0.97) and Chu et al. (2013a), (2013b). In the latter study, they were also positively correlated with the HCR-20: Assessing Risk for Violence clinical subscale. All three tools were significantly correlated with proxy clinical risk measures included in studies of their psychometric properties. Woods et al. (2015) reported the Brøset Violence Checklist to be positively associated with a visual clinical judgement slide rule (r_{pb}=0.57) while Abderhalden et al. (2004) reported Spearman's rho correlations between a 10-centimetre Visual Analogue Scale and the Brøset Violence Checklist total score of 0.5 and 0.59 in tool development and tool validation studies respectively. In Dumais et al. (2012), Dynamic Appraisal of Situational Aggression clinical judgement was positively associated with a clinical judgement rating of risk (Pearson $r = 0.75$). Sarver et al. (2019) found that patients with longer length of stay and those requiring high level of restrictive interventions for violence during their inpatient stay had significantly higher admission Brøset Violence Checklist scores than their counterparts.

3.2.5. Tool modifications

In a two-part study, Abderhalden et al. (2006) supplemented the Brøset Violence Checklist total score with a 10-cm visual analogue scale on which assessors used clinical judgement to estimate

the likelihood of the patient being physically assaultive (study one). From the resulting analysis, a slide-rule tool was developed which converted scale ratings into a score reflecting the actuarial probability of an attack (e.g., 'at this level of risk 1 in 10 patients will attack someone'). In study two, conducted with a different sample, predictive validity was calculated for the Brøset plus slide rule rating. In study one the visual analogue scale rating (AUC =0.94) exceeded the raw Brøset score alone (AUC=0.87) while a combined, equally weighted visual analogue scale plus Brøset score achieved AUC=0.95. In study two the Brøset plus slide rule score (AUC=0.83) did not exceed the Brøset alone (AUC=0.86) for prediction of physical assault; for prediction of physical assault or intense physical intervention comparison of AUC CIs revealed non-significant differences (AUC = 0.90. 0.88, and 0.85 for the combined Brøset plus slide rule, Brøset alone, and slide rule alone respectively). AUC confidence intervals suggest no significant differences between any comparison pairing.

Loi and Marlowe (2017) reported on a modified Brøset Violence Checklist (additional items: response to de-escalation and medication compliance) designed to retrospectively audit decision-making about seclusion. They reported an AUC of 0.84 and of 0.94 for prediction of the use of seclusion for the six original items and for the modified tool respectively.

Riordan et al. (2019) have supplemented the Dynamic Appraisal of Situational Aggression with female-specific items, but this did not result in more accurate prediction of physical aggression towards others (original tool AUC= 0.62 [95% CI 0.59, 0.66] vs. supplemented womens' version (AUC= 0.63 [95% CI 0.59, 0.66). Lansgrud et al. (2019) considered the effect of an item representing quality of sleep as an addition to the Brøset Violence Checklist assessment and calculated its effect on the predictive value of the tool. While an improved AUC of 0.873 [0.75, 0.99] (compared with 0.757 [0.61, 0.91]) was achieved, overlapping 95% CIs suggested this was not statistically significant.

3.2.6. Non-standard rating periods

Most studies have calculated the predictive performance of the Brøset Violence Checklist or Dynamic Appraisal of Situational Aggression for the 24-h period following assessment. However, instructions for both tools indicate that rating should take place each nursing shift, and some studies have calculated predictive validity of rating over this shorter period. Direct comparison has resulted in superior values for 8-h versus 24-h (Almvik et al., 2007 [Brøset Violence Checklist]; Dumais et al., 2012 [Dynamic Appraisal of Situational Aggression]). In Almvik et al. (2007), performance over 8-h was exceptionally potent (AUC=0.98) compared with 24-h (AUC=0.69). In addition, one study (Nqwaku et al., 2018) has examined prediction of both peak and average Dynamic Appraisal of Situational Aggression scores over a 24-h period, compared with the single assessment immediately prior to follow-up. The most potent predictive combination was average scores from the preceding 24-h period coupled with a follow-up period of 24-h; 8-h follow up achieved smaller AUCs.

3.2.7. Use as an intervention

Brøset violence checklist: In Abderhalden et al.'s cluster randomised controlled trial on 14 acute psychiatric admission wards, nine wards applied the Swiss language version of the Brøset Violence Checklist augmented by the slide-rule visual analogue scale twice daily for the first 3-days of hospital admission. This resulted in reduction of physical attacks of 41% and 52% in the intervention wards (randomised to and opted in wards respectively) but only 15% in wards assigned to a control arm. Coercive measures reduced by 27% and 60% in the intervention wards but only 10% in control wards. The trial confirmed an earlier report of a study conducted on two wards only (Needham et al., 2004). Van de Sande et al.'s (2011) cluster randomised controlled trial in four wards over a 40-week period saw two wards assigned to routine Brøset Violence Checklist use and two to clinical judgement alone. The number of aggressive incidents reduced significantly on Brøset-assigned wards (4.9–1.7 incidents per week) but not clinical judgement wards (3.5 per week to 3.9 per week) after controlling for number of patient days. The amount of time spent in seclusion, but not the actual number of seclusion events, reduced significantly in Brøset wards. In contrast, Hvidhjelm et al. (2016) investigated the effect of the introduction of the Brøset Violence Checklist in seven intervention wards compared with care as usual in eight control wards. While the risk of aggression was reduced by 45% in intervention wards it was not at a level of statistical significance. Two studies (Clarke et al., 2010; Blair et al., 2017) reported pretest-posttest evaluations of Brøset Violence Checklist implementation. Clarke et al. (2010) reported seclusion rates falling from 30 to 12 per month over the implementation period, and a subsequent rise to 22 per month after the formal intervention period. Blair et al. (2017) describe a quality and safety initiative in which an intervention, including routine Brøset Violence Checklist assessment, was introduced into a 120-bed psychiatric service. Compared to baseline data from the year prior to the initiative, the rate of seclusion fell from 9.2 to 4.4 per 100 admissions. However, some unwanted changes such as increases in duration of restraint were also revealed.

Dynamic appraisal of situational aggression: Formal trials have not been conducted. In an evaluation of the New Zealand-based High Risk Personality Programme–Revised, a comprehensive rehabilitation programme for maximum security prisoners, Wilson and Kilgour (2015) state that daily assessment provided 'an *objective and independent measure of demonstrable behaviour change*' (Section 'Incidents and misconducts' Para 1). The authors report a near-70% reduction in the occurrence of violent incidents over the course of the programme but the proportion attributable to Dynamic Appraisal of Situational Aggression implementation, if any, is unclear.

Kaunomäki (2015) and Kaunomäki et al. (2017) examined the occurrence of interventions following violence risk assessment and their subsequent relationship with change in score the following day. Results demonstrated that the next day's Dynamic Appraisal of Situational Aggression sum score decreased where the patient received one to three interventions following the previous day's assessment. Four or more interventions had no discernible statistical relationship with the subsequent Dynamic Appraisal of Situational Aggression score. Common interventions were pro re nata medication, seclusion and focused discussions with a nurse. Non-coercive and non-pharmacological interventions including daily activities were most significantly linked with decreased perceived risk of violence.

3.2.8. Predictive validity

Ogloff and Daffern (2006a) reported that the proportion of people who displayed each Dynamic Appraisal of Situational Aggression behaviour who went on to be violent ranged from 29.5% to 56.8% (Mdn=38.6%). The most discriminating items, indicated by the greatest difference between presence and absence among violent and non-violent patients, were irritability, verbal threats, and impulsivity. Almvik et al. (2000) reported significant differences in the proportions of violent and non-violent patients with the presence or absence of positive ratings for all 6 Brøset Violence Checklist items. Most notably, attacks on objects (32.8% of violent patients, 1.1% of non-violent patients) and boisterousness (53% of violent patients, 5.4% of non-violent patients) clearly discriminated between violent and non-violent patients. Sarver et al.'s study of admission Brøset Violence Checklist ratings of 222 adult psychiatric inpatients found that each additional point scored was associated with a 3.4 times increase in violence on day one of their

stay. In Björkdahl et al. (2006) Brøset Violence Checklist study, the most potent predictor-items were irritability and boisterousness. In Almvik et al. (2007)'s Brøset Violence Checklist study among older people, irritability was also the most commonly rated item among violent subjects (82.8%), followed by confusion (48.3%) and verbally threatening behaviour (46.1%). In terms of item performance, Chu et al. (2012) found that only the Dynamic Appraisal of Situational Aggression items 'negative attitudes' and 'unwillingness to follow directions' were significantly predictive of violence in the 48-h following assessment. However, Dutch and Patil (2018) found that nine of the eleven Dynamic Appraisal of Situational Aggression Youth Version items produced significant AUCs for prediction of aggression, the exceptions being 'anxious/fearful' and 'outside stressors'.

For the Dynamic Appraisal of Situational Aggression, all but four studies reported an outcome category of "any aggression." The exceptions were Dumais et al. (2012) who did not report on verbal aggression either separately or as part of a compound 'all aggression' outcome; and Ogloff and Daffern (2006a) who reported on more serious physical aggression against people only. Lantta et al. (2016) and Vojt et al. (2010) did not report an 'any aggression' outcome but in these two cases, the mean of all non-overlapping effect sizes related to aggression was coded into an "any aggression" category. Eight studies using the Dynamic Appraisal of Situational Aggression reported information sufficient to allow calculation and subsequent analysis of pooled effect sizes for 'physical aggression' and five presented sufficient information about verbal aggression for analysis. For the Brøset Violence Checklist, eight studies reported an "any aggression" outcome, and six a "physical aggression against people" outcome; for only one sample (Chu et al., 2013a, 2013b) were both reported. In contrast, both outcomes were reported for six of twelve samples assessed with the Dynamic Appraisal of Situational Aggression.

Across studies there were 78 unique total score–24-h outcome dyad combinations of which 75 (96.2%) performed significantly better than chance, the exceptions being the Dynamic Appraisal of Situational Aggression total score – physical aggression against people combination in Chu et al. (2012), and the physical aggression against staff and patient combinations reported by Vojt et al. (2010). After collapsing categories to their final versions (see 2.5.1) 72 unique predictor-outcome combination AUCs, reported here as Hedges' g values, remained; 13 (18.3%) were present in only one study and the remaining 59 combinations related to five outcomes (any aggression, physical aggression against people, physical aggression against objects, verbal aggression, and self-harm; range 2–11) and four tools (range 1–4). For all 11 studies with a Dynamic Appraisal of Situational Aggression total predictor and an 'any aggression' outcome pairing, the former significantly predicted the latter (g range 0.32–2.66, Mdn=0.91); similarly, all six studies presenting comparable information for the Brøset Violence Checklist predicted at significantly greater than chance levels (g range 0.85–1.71, Mdn 1.25). For physical aggression against people, seven of eight Dynamic Appraisal of Situational Aggression studies indicated statistically significant prediction (g range 0.29–1.97, Mdn=0.74); while for the Brøset Violence Checklist, all six results were predictive (g range 1.29–1.99; Mdn=1.47). Verbal aggression was significantly predicted by all six studies presenting results for the Dynamic Appraisal of Situational Aggression (g range 0.40–1.53, Mdn=0.82).

3.2.9. Meta-analyses of predictive validity

Prior to outlier detection, examination of assessment - outcome pairings revealed large effect sizes (g >0.8) for both the Brøset Violence Checklist and the Dynamic Appraisal of Situational Aggression in relation to physical aggression ($k=8$, $n=163{,}563$; $k=8$, $n=37{,}569$), and any aggression ($k=6$, $n=17{,}889$; $k=11$, $n=52{,}896$) in both random and fixed effects models. In relation to verbal aggression the Dynamic Appraisal of Situational Aggression analysis also revealed a large effect size ($k=5$, $n=19{,}288$). For the Dynamic Appraisal of Situational Aggression, Youth Version where no outlier analysis or moderator analyses were conducted due to the small number of studies in the analyses ($k=2$, $n=4905$), pooled effect size (random effects model) for prediction of any aggression was $g=1.39$ [95% CI 0.57, 2.21]; for physical aggression it was $g=1.24$ [95% CI 0.42, 2.06]. For the HCR-20: Assessing Risk for Violence clinical subscale, which similarly was not subject to outlier or moderator analyses, prediction ($k=2$, $n=13{,}492$) of physical aggression produced a large pooled effects size ($g=0.86$, 95% CI 0.82, 0.89).

3.2.10. Outlier and moderator analyses

Outlier analyses revealed that one study involved in the Dynamic Appraisal of Situational Aggression –Any Aggression analysis (Chan and Chow, 2014) was an outlier since Q (d.f.=10) reduced from 24239.35 to 1250.65 (d.f.=9) following removal. Removal resulted in reduction of random and fixed pooled effect sizes from 2.22 and 1.04 to 0.88 and 0.87 respectively ($k=10$, $n=45{,}537$). I^2 remained very high (>99%) both with and without the outlier. Moderator analysis revealed, however, that heterogeneity was not significantly contributed to by any of the variables tested. The Dynamic Appraisal of Situational Aggression – Physical Aggression analysis revealed no outliers. Pooled effect size was large under both random ($g=1.04$ [0.69, 1.38]) and fixed effects ($g=0.95$ [0.91, 0.98]) models (see Fig. 2). Moderator effect was significant (Q [d.f.=7]=277.18, $p<.0001$); significant moderators were: study bias, where studies at low risk of bias produced significantly smaller effect sizes compared to studies at high risk of bias (−1.25 [−1.73,−0.77]); legal status, where studies of forensic (0.68 [0.18, 1.17] $p<.05$) and mixed forensic/civil patients (0.88 [0.40, 1.36] $p<.001$) produced greater effect sizes compared to civil patients; and the number of observations per participant, where effect sizes decreased by 0.0064 [−0.009, −0.003] for every additional observation per patient. A high amount of heterogeneity remained in the data after accounting for moderators (I^2=56.76%%). For the Dynamic Appraisal of Situational Aggression – Verbal Aggression analysis, Lantta et al. (2016) met criteria for outlier status ($Q=109.13$ pre-removal, $Q=47.60$ post removal, $k=5$, $n=19{,}110$). The associated reduction in I^2 was, however, small (96.3–92.2%). Moderator analysis revealed significant effects of low risk of study bias (−0.98 [−1.75, −0.22]).

For the Brøset Violence Checklist – Any Aggression analysis, the study by Almvik et al. (2007) met criteria for outlier status ($Q=968.48$ pre-removal, $Q=119.18$ post-removal), but removal only led to a marginal reduction in between-study heterogeneity (I^2=98.82% to I^2=95.88%). Removal of the study ($k=7$, $n=8993$) resulted in pooled effect size increase from $g=1.13$ [0.82, 1.44] to 1.25 [0.99, 1.50]. Moderator analyses revealed no significant effect for any of the moderators tested. For the Brøset Violence Checklist –Physical Aggression analysis (see Fig. 3), the study of Hvidhjelm et al. (2014) met the criteria for outlier status with removal reducing in lowered pooled effect size ($k=7$, $n=24{,}074$). Random ($g=1.34$ [1.15, 1.56]) and fixed ($g=1.28$ [1.26, 1.31]) effects models both produced large pooled effect sizes with significant heterogeneity ($Q=238.21$, d.f.=5, $p<.0001$; $I^2=97.9\%$). Moderator analyses revealed significant effects of low study bias (0.16, 95% CI 0.02, 0.31, $p<.05$), proportion of males in the sample (−0.06 95% CI −0.07, −0.05, $p<.0001$), forensic status (0.68 95% CI 0.47, 0.88), and violence base rate (−0.42, 95% CI −0.49, −0.35). Only one study (Chu et al., 2013a, 2013b) used the Brøset Violence Checklist to predict verbal aggression ($g=1.04$ [0.89, 1.20]) and thus no inspection of heterogeneity or moderator analysis applied.

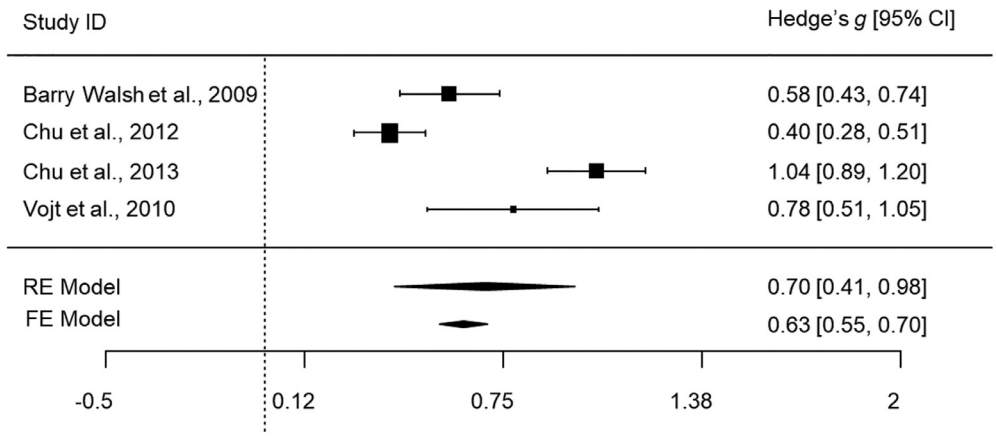

Fig. 2. Forest plot showing individual and pooled effect sizes for Dynamic Appraisal of Situational Aggression prediction of physical aggression.

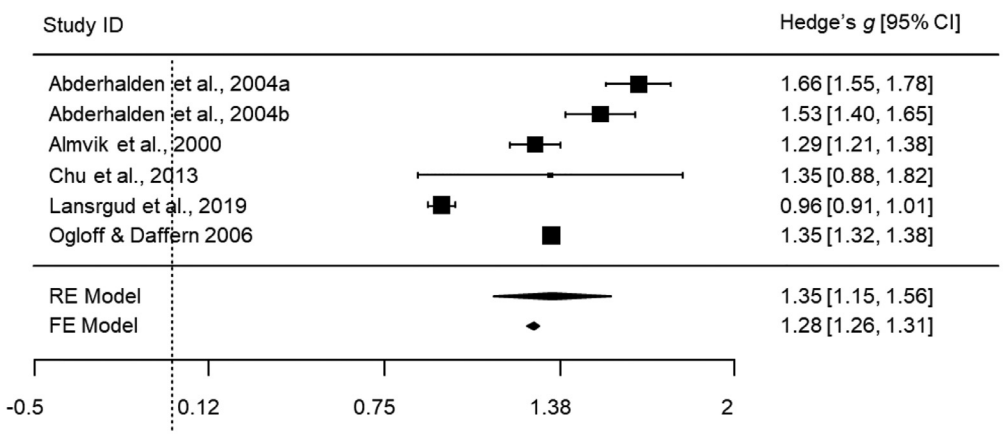

Fig. 3. Forest plot showing individual and pooled effect sizes for Brøset Violence Checklist prediction of physical aggression (Hvidhjelm et al., 2014 removed from analysis as an outlier).

3.2.11. Differential predictive validity

Chu et al. (2012) presented AUCs for males ($n = 29$) and females ($n = 20$) separately. While there had been significant predictive efficacy of the Dynamic Appraisal of Situational Aggression total score for verbal aggression among the whole sample (AUC=0.59 (95% CI 0.49, 0.73), this effect remained only for boys in subsequent subgroup analysis (AUC=0.62 95% CI 0.50, 0.73) which may reflect the lower base rate (20.0% vs. 44.8% for males) and small sample of females in the study. Maguire et al. (2017) found no gender differences in terms of predictive accuracy; however, Dynamic Appraisal of Situational Aggression score was significantly predictive of aggression by both men and women. Almvik et al. (2007) presented subgroup analyses for general psychogeriatric patients and for those with dementia; performance was significant in both groups, stronger in the former (AUCs 0.76 vs. 0.70) but the difference was not statistically significant. Wood et al.'s (2015) subgroup analyses of predictive performance for patients with dementia, psychosis, or substance use disorder also suggested significant performance for the tool in all subgroups (AUCs 0.67, 0.81 and 0.76, respectively). However, confidence intervals indicated that performance was not significantly different between subgroups.

3.2.12. Incremental validity

There has been no formal investigation of the incremental validity of the Brøset Violence Checklist over any other method of prediction. For the Dynamic Appraisal of Situational Aggression, Chu et al. (2013a, 2013b) asked assessors to rate risk as 'low', 'medium' or 'high' to supplement total score; analysis revealed that the total score had significant incremental predictive validity over these clinical judgements.

3.2.13. Sensitivity

The ability of either the Dynamic Appraisal of Situational Aggression or Brøset Violence Checklist to track patient change over time has not been studied, though rare case reports exist where assessments have been used to illustrate an individual patient/prisoner's progress (Tucker et al., 2016; Wilson and Kilgour, 2015).

4. Discussion

We have systematically identified and appraised instruments for risk assessment of imminent violence, defined as 24-h from assessment, in residential psychiatric and correctional facilities. Five tools have been used for this purpose: four items are shared by three tools each because the Dynamic Appraisal of Situational Aggression and its variants include two items each from the Brøset Violence Checklist and the HCR-20: Assessing Risk for Violence clinical subscale. Meta-analyses of the Brøset Violence Checklist and Dynamic Appraisal of Situational Aggression revealed large pooled effect sizes for the prediction of categories of 'any aggression' and 'physical aggression against people' before and after outlier analysis.

Due to the small number of studies available for inclusion in the meta-analyses, only a limited number of moderators could be investigated leaving a wide range of potentially important variables uninvestigated in the current study. Previous systematic reviews of the role of moderators in inpatient violence prediction studies include examination of the role of clinical diagnosis (e.g., proportion of sample with schizophrenia or personality disorder), ethnicity, age, and length of follow-up (O'Shea et al., 2013). Other variables shown to be of potential importance in individual Brøset Violence Checklist and Dynamic Appraisal of Situational Aggression studies include hospital-site, stage of admission (e.g., prediction in the early stages of admission versus across the whole admission), and nurse perception of impairment due to recent drug misuse (Palmstierna and Olsson, 2007). Additionally, we can reasonably speculate about the role of ward climate, organisational culture, staff attitudes and knowledge, and nursing skills in moderating the relationship between risk assessment and predictive accuracy.

Outlier analysis revealed the Brøset Violence Checklist pooled effect size for any aggression prediction to be unduly influenced downwards by the inclusion of Almvik et al. (2007) study of older age adults including those with dementia. This makes clinical sense since it is reasonable to expect that prediction will worsen for a group so significantly dissimilar from validation samples in terms of neurodegeneration. Between-study variance in Brøset Violence Checklist any aggression performance was unexplained by the moderator variables studied. For prediction of physical violence a very large effect size was present even following removal of an outlier (Hvidhjelm et al., 2014). Moderators of significance included forensic legal status (larger effect sizes), male gender proportion (smaller effect size) and study bias (greater effect sizes for studies at risk).

Outlier analysis of the Dynamic Appraisal of Situational Aggression revealed an undue influence from the sample contributed by Chan and Chow (2014); removal resulted in a fall in pooled effect size for prediction of any aggression but retention of a large effect size. Between-study variance was partly explicable by moderators for 'physical aggression against people': similarly to the Brøset Violence Checklist, effect sizes were significantly inflated by increased study bias, and the forensic status of study participants. Additionally, Dynamic Appraisal of Situational Aggression studies with larger numbers of assessments per participant resulted in lower effect sizes.

In total, the performance of both tools was broadly similar; head-to-head, Brøset Violence Checklist pooled effect sizes were greater than the Dynamic Appraisal of Situational Aggression however, overlapping 95% CIs indicated no statistically significantly superior performance. The Dynamic Appraisal of Situational Aggression youth and women's versions are currently less well evidenced in terms of all psychometric properties but surely must share some qualities of the parent tool with which they share 7-items. The HCR-20: Assessing Risk for Violence clinical subscale was included in early validation studies of the Dynamic Appraisal of Situational Aggression where two of its items were found to add predictive validity to constellations of other items and were thus adopted into the newer tool. Despite a lack of empirical evidence for the psychometric properties of the HCR-20: Assessing Risk for Violence clinical subscale for use specifically as an assessment for imminent risk, there is an abundance of evidence supporting its use in longer-term risk assessment (e.g., Ramesh et al., 2018; Singh et al., 2014).

Our results confirm findings of a previous analysis (Ramesh et al., 2018), that risk assessment tools for prediction of imminent aggression in forensic settings produce results significantly better than chance. Our paper, which includes a broader range of studies conducted in multiple, rather than solely forensic, settings, extends this finding to civil and mixed hospitals with the caveat of the moderating influence of participant forensic status noted above. In addition, the moderator analysis is an extension of previous work. However, the analysis did reveal only limited significant effects of the variables chosen for inclusion. This means that the considerable heterogeneity between studies remains unexplained for the large part.

The current review is also distinguished by a thorough narrative analysis of the properties of tools that have been used in imminent violence risk research. The review suggests the tools are generally acceptable to, and easy to use by, practitioners, most commonly nurses. There was a dearth of information about interrater reliability particularly under ecologically valid (i.e., non-vignette type) conditions. Given that this is a likely source of between-study heterogeneity – for example, items subject to subtly different interpretations between settings – it is necessary to conduct further research. It is likely that Dynamic Appraisal of Situational Aggression and Brøset Violence Checklist studies include far larger numbers of raters (i.e., all nurses on a team) than in those of tools for medium- and longer- term violence prediction which are commonly researcher-rated from clinical records, an approach only conducted, to our knowledge, in Loi and Marlowe's (2017) study which provided little detail about its equivalence to routine rating. It was perhaps surprising that only one dedicated implementation study has been conducted. Lantta's (2015) and Lantta et al. (2015, 2016) accounts suggest there is value to be added to further empirical studies of imminent-aggression risk instruments through the application of techniques from implementation science which aim to answer questions about how best to translate research into practice (Bauer et al., 2015).

Attempts to otherwise modify the tools to improve their performance have been largely unvindicated. Brøset Violence Checklist performance was not improved by augmentation with a visual analogue scale (Abderhalden et al., 2004); nor that of the Dynamic Appraisal of Situational Aggression when augmented with additional 'female-friendly' items (Riordan et al., 2019). Studies examining differential predictive validity have been rare. No gender differences were reported in studies by Chu et al. (2012), and no diagnostic differences in Almvik et al. (2007) study. At present, the few development studies of the Dynamic Appraisal of Situational Aggression youth and women's variations have indicated improved predictive validity over the original tool but there is little to indicate that it will be fruitful to develop multiple gender-, diagnosis-, or other-specific versions of tools if prediction is the sole outcome of interest. Especially in mixed clinical settings this would likely complicate rating with little in the way of gain in accuracy. However, previous studies have found that clinicians find risk assessment schemes to have clinical utility beyond risk prediction (Nicholls et al., 2011), and it may be that extended versions offer practical utility to risk assessment.

Related to modified tools is the emergence of studies about the use of machine learning techniques for risk prediction. Van Le et al. (2018) have tested multiple machine-based dictionaries' ability to examine natural language occurring in clinical notes and use this to predict clinicians' own risk assessment judgements including on the Dynamic Appraisal of Situational Aggression. Hotzy et al. (2018) have reported prediction of actual patient outcomes equivalent to traditional methods following 'training' of machines on previously identified risk factors for being subjected to coercive practices. One can immediately envision the financial benefits and the potential ethical dangers of automated processes in the conduct of high volume, high frequency tasks such as imminent violence risk assessment. Great attention should be paid as these methods develop in the coming years.

Data analyses have extended to determining the optimum ratings and outcomes periods. There is some evidence that accuracy of the Brøset Violence Checklist improved when the follow-up was

8-h rather than the standard 24-h period (Almvik et al., 2007; Dumais et al., 2012); and this aligns with the original intended frequency of rating recommended by the tools' authors though results were not statistically significant. Similarly, Nqwaku et al. (2018) reported that a composite Dynamic Appraisal of Situational Aggression rating, derived from the ratings across the 24-h period prior to assessment, improved predictive accuracy significantly. However, there is clearly a balance to be struck between the predictive value added by rating per shift and using composite measures and the commensurate drain on resources in terms of nursing time. There has been no serious attempt to examine the sensitivity of either the Dynamic Appraisal of Situational Aggression or Brøset Violence Checklist to change in risk over time.

Work to date suggests limited but promising evidence that implementation of the Brøset Violence Checklist can significantly reduce aggression with Abderhalden et al. (2008), Needham et al. (2004), and Van de Sande et al. (2011) all reporting positive results in controlled trials. No similar studies of the Dynamic Appraisal of Situational Aggression have been conducted.

Inevitably, given the availability of only a small range of tools for analysis, clinicians and researchers might reasonably ask which is the superior instrument? There is little to choose between the two; the most superior effect sizes were produced for prediction of physical aggression by the Brøset Violence Checklist, but that for any aggression is equivalent and, in any event, differences were not statistically significant. It seems clear that use of either tool has the potential to deliver clinical advantages in both forensic and civil settings and across patient-groups. Relevant considerations in selecting a scheme for implementation will include aspects encompassing ease of use, time taken, clarity of coding manuals, necessary training time and cost, relevance of development and validation data to the clinical setting, and likely clinical utility.

4.1. Limitations

We have defined 'imminent' for the purposes of the current review as the 24-h period following aggression. Our selection was pragmatic, but essentially arbitrary based on the frequency of research conducted across this time period. There is no universally accepted definition of imminent, and ours may be considered a limitation in terms of its ability to detect all relevant studies. One survey identified over 400 different violence risk assessments in use worldwide (Singh et al., 2014) and it is possible that our search did not identify all of these since many were locally developed and unpublished. The predictive validity of tools not included in the present study has been assessed over the course of admission; in a sense, then, they may fit our criteria because some of the incidents involved in studies of predictive validity may occur during 24-h. However, this is true of all violence risk assessment instruments and we are not aware of any analyses which have examined, for instance, survival rates (time to first violent act) separately for those incidents occurring 'imminently'. In our estimation, the dictionary definition of 'imminent' as "about to happen" or "at any moment" is more congruent with the shortest time periods. The developers of both the Brøset Violence Checklist and the Dynamic Appraisal of Situational Aggression recommend rating three times per day which also supports this interpretation. We therefore believe that the review covers the entirety of empirical research about a discrete section of the violence risk assessment literature.

The study is limited by the small range and diversity of imminent aggression instruments available, and by the quality of those studies. The greatest potential limitation is the likely bias in studies arising from doubtful inter-rater reliability, this being the item most implicated in the quality ratings of studies. Questions also remain about the suitability of these tools for analysis using methods such as the receiver operating characteristic. The data sets inevitably comprise multiple ratings, sometimes hundreds or thousands per patient, a situation not normally found in violence risk assessment studies where the prediction period can be months or years. Recently, Riordan et al. (2019) have analysed data from 13 patients measured multiple times and, after controlling for this, report AUC values of up to 0.92 versus 0.62 prior to control. Ramesh et al. (2018) previous review eschews analysis of the AUC of the receiver operating curve precisely for this reason, depending instead on dichotomised raw data sourced from researchers and binning patients into violent/non-violent categories. It seems important that researchers of tools like the Brøset Violence Checklist and Dynamic Appraisal of Situational Aggression agree on a common reporting method and make collaborative efforts to investigate the best statistical methods to deal with multiple measures. Creative studies of inter-rater reliability and other important implementation questions are required as a matter of urgency if we are to continue to claim these tools have a well-established evidence base.

4.2. Conclusions

Structured instruments for the prediction of imminent aggression are a relatively new addition to the clinicians' basket of tools to describe, predict, and prevent violence in psychiatric and correctional populations. Ultimately, the point of risk assessment is to identify target causal risks so that they can be treated or otherwise ameliorated, thus improving safety for staff, and increasing the likelihood of successful rehabilitation and reintegration of the patient or prisoner into society. Structured risk assessment for imminent violence is promising and worthwhile; either of the most well-used tools – the Brøset Violence Checklist or the Dynamic Appraisal of Situational Aggression – might be implemented and add value to clinical practice in either forensic or civil settings. Concurrently, the complexity of introducing these tools for use by large teams of nurses must be further investigated and subject to development if their value is to be maximised. We recommend the adoption of a definition of 'imminent risk' as the 24-h period following assessment.

Conflict of Interest

All authors declare that they have no conflicts of interest

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